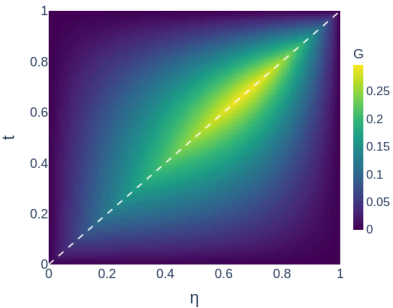


Numerical Evidence I: GreenNet Kernel Approximation

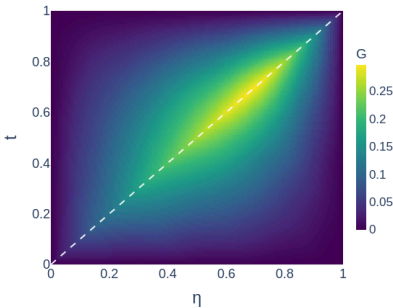
Capturing the singular Green structure on an axial interval

$$\Omega = \{x^2 + y^2 < 0.5^2\}, \quad -\nabla \cdot (a \nabla u) = f, \quad a(x, y) = 1 + \frac{1}{2} \sin(2\pi x) \sin(4\pi y), \quad \mathbf{b} = 0, \quad c = 0.$$

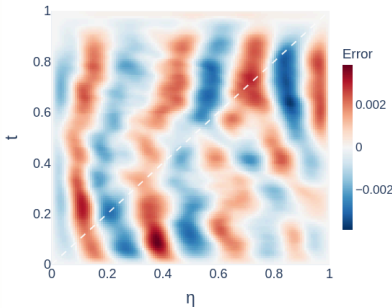
REFERENCE KERNEL



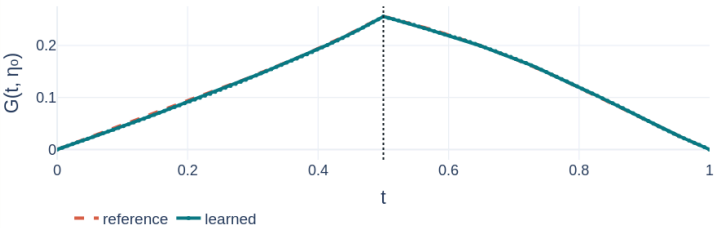
LEARNED KERNEL



SIGNED ERROR



FIXED-H SLICE



DIAGNOSTICS

η	0.50
Kernel rel. L2	1.11%
Slice rel. L2	0.92%
Diagonal band	$ \mathbf{t}-\boldsymbol{\eta} \leq 5/128 = 0.0391$
Error mass / band area	9.5% / 8.3%
Mean error / off-band mean	1.16x

The learned kernel captures the singular Green structure, and the signed error is not concentrated near the singular diagonal